Mohammad Saadati

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Summary

M.S. student in artificial intelligence, currently working on ARC-AGI-3 with a focus on reinforcement learning and large language models. Experienced software engineer with a track record of building production-ready systems.

Education

Gwangju Institute of Science and Technology (GIST) Gwangju, South Korea
M.S. in Artificial Intelligence; GPA: 4/4.5 *Sep. 2025 - Present*
Advisor: *Prof. Sundong Kim*
Relevant Courses: *Robot Reinforcement Learning, Computer Vision, Generative AI Blockchain, Advanced Deep Learning, Intelligent Agent, Autonomous Driving*

University of Tehran Tehran, Iran
B.Sc. in Computer Engineering *Sep. 2019 - Jul. 2024*
Relevant Courses: *Reinforcement Learning(Master's course), Advanced Interactive Learning(Master's course), Artificial Intelligence, Data Mining(Master's course), Game Theory, Engineering Probability and Statistics, Data Structure and Algorithm, Discrete Mathematics, Advanced Programming, Operating Systems, Database Design, Software Engineering, Internet Engineering, Distributed Computing, Operational Research*

Research Interests

- Reinforcement Learning
- Machine Learning
- Data Science & Big Data
- Multi-Agent Systems & Robotics
- Large Language Models
- Software Engineering

Research Experience

Graduate Research Assistant DataScience Lab, GIST, South Korea
Under the supervision of Prof. Sundong Kim *Sep. 2025 - Present*

- Working on **ARC-AGI-3** to achieve state-of-the-art performance under offline evaluation constraints, where no external supervisor is available at inference time. Developing an agentic AI system that derives supervisory signals purely from observations to drive IDA* search, reinforcement learning, and case-based memory, enhancing agent decision-making and generalization across ARC-AGI-3 environments. Additionally, investigating the use of large language models to synthesize executable world models from interaction traces for downstream planning.

Undergraduate Research Assistant University of Tehran, Iran
Under the supervision of Prof. Majid Nili *Jan. 2023 - Jul. 2024*

- Explored the application of **transfer learning** to improve the efficiency of **reinforcement learning** by leveraging **large language models**. The research focused on using these models to identify similarities and differences between previous knowledge and new tasks, facilitating more effective learning transfer. Experiments demonstrated that large language models could significantly reduce the learning time and enhance decision-making processes, leading to improved performance in reinforcement learning scenarios. [Poster]

Undergraduate Research Assistant University of Tehran, Iran
Under the supervision of Prof. Azadeh Shakery *Jul. 2022 - Jun. 2023*

- Developed a **hate speech detection** model by integrating **GCN (Graph Convolutional Networks)** with **BERT**, utilizing a heterogeneous graph for document classification. The research, informed by a thorough literature review, successfully optimized graph relationships to improve accuracy.

Undergraduate Research Assistant University of Tehran, Iran
Under the supervision of Prof. Behnam Bahrak *Aug. 2021 - Dec. 2022*

- Created a hybrid **consensus protocol** combining **proof of stake (PoS)** and **proof of activity (PoA)**. This work, grounded in an extensive study of similar protocols, effectively leveraged the benefits of both PoS and PoA.

Publications

- **Mohammad Saadati**; Laura Mascarell; and Sundong Kim. **"A Non-LLM Graph-Search Agent for ARC-AGI-3 with Novelty-Guided Exploration and Case-Based Memory."** Korea Computer Congress (KCC), 2026. [Poster]
- Sina Kamali; Shayan Shabihi; Mohammad Taha Fakharian; Alireza Arbabi; Pouriya Tajmehrabi; **Mohammad Saadati**; and Behnam Bahrak. **"RPoA: Redefined Proof of Activity."** arXiv preprint arXiv:2210.08923, 2022.

Work Experience

Aban Prime Co. Tehran, Iran

Software Engineer *Apr. 2025 - Aug. 2025*

- Contributed to a dynamic crypto exchange startup by taking on diverse responsibilities across backend development and process optimization.
- Developed a notification center using **ASP.NET, PostgreSQL, and unit testing** to enhance user engagement and streamline communication within the platform.
- Improved the IRR banking deposit process by leveraging **artificial intelligence (AI) with Django, Celery, Redis, and Docker** to simplify account charging for support teams.
- Automated banking information delivery for IRR and AED deposits to overcome various banking account limitations.

Pars Jahd Green Technologies Co. Tehran, Iran

Software Engineer *Mar. 2024 - Mar. 2025*

- Implemented a web-based platform for securely managing personal health records. Utilized **Python** and the **Django** framework to develop the core functionalities and advanced features, integrating **MySQL** and **PostgreSQL** databases. Employed tools such as **Celery, unit testing, Docker, CI/CD pipelines,** and **OAuth** for secure authentication. My work ensures seamless data management, enhances doctor-patient interactions, and supports accurate diagnoses through comprehensive record-keeping. [Platform Website]

Teaching Assistant Experience

University of Tehran

- **Data Science** *Prof. B. Bahrak, Prof. Y. Yaghoobzadeh*s Spring 2024
- **Reinforcement Learning (Master's course)** *Prof. M. Nili* Fall 2023
- **Artificial Intelligence** *Prof. Y. Yaghoobzadeh, Prof. H. Fadaei* Spring 2023, Fall 2023
- **Data Mining (Master's course)** *Prof. A. Shakery* Spring 2023
- **Operating Systems** *Prof. M. Kargahi* Fall 2022, Spring 2023, Fall 2023
- **Discrete Mathematics** *Prof. S. Mohammadi* Spring 2022, Fall 2022, Spring 2023
- **Engineering Probability and Statistics** *Prof. B. Bahrak* Fall 2022
- **Introduction to Computing Systems and Programming** *Prof. H. Moradi* Fall 2020, Fall 2021

Honors & Awards

- Received Korean Government Scholarship, Gwangju Institute of Science and Technology (GIST) 2025
- Received scholarship, Supporter Foundation of the University of Tehran 2019
- Ranked Top 0.2% in Konkour, Iranian University Entrance Exam among 150666 participants 2019

Coursera Certifications

- Reinforcement Learning Specialization
- Natural Language Processing Specialization
- Applied Text Mining in Python
- Text Retrieval and Search Engines
- Text Mining and Analytics
- Machine Learning Specialization
- Deep Learning Specialization
- Mathematics for Machine Learning Specialization
- Statistical Inference
- Generative Adversarial Networks (GANs) Specialization
- Machine Learning Engineering for Production (MLOps) Specialization
- Practical Data Science on the AWS Cloud Specialization
- Google Data Analytics Professional Certificate

Professional Developments

Crystaline University of Tehran, Iran

A cryptocurrency Powered by a Redefined POA Protocol *Aug. 2021 - Sept. 2022*

- Developed as a proof of concept on pure Python, this **cryptocurrency** incorporates a newly defined **proof of activity (PoA)** as its primary **consensus protocol**. It was designed and researched by me and several other students at the University of Tehran.

Skills

Programming: Python, C++, C, C#, Java, Go, SQL, Verilog, R
Frameworks & Libraries: Django, ASP.NET, NumPy, Pandas, Scikit-Learn, TensorFlow, Keras, PyTorch, NLTK
Tools & DB: Git, LaTeX, Makefile, Jupyter, Docker, Kubernetes, Maven, JUnit, MySQL, PostgreSQL, MongoDB, Redis
Software Engineering: Design Patterns, Object-Oriented Design, Agile, Scrum, CI/CD
Web Development (Familiar with): HTML, CSS, JavaScript, Bootstrap, Tomcat, Spring, React
Operating System: Linux(Ubuntu), Windows
Languages: Persian(Native), English(Fluent) - IELTS Academic: Overall 6.5 [L:6, R:7, W:6.5, S:6] (Sep. 2023)
Interests & Hobbies: soccer, fitness, music, computer games, movies, traveling.